

Nibrexo OS AI

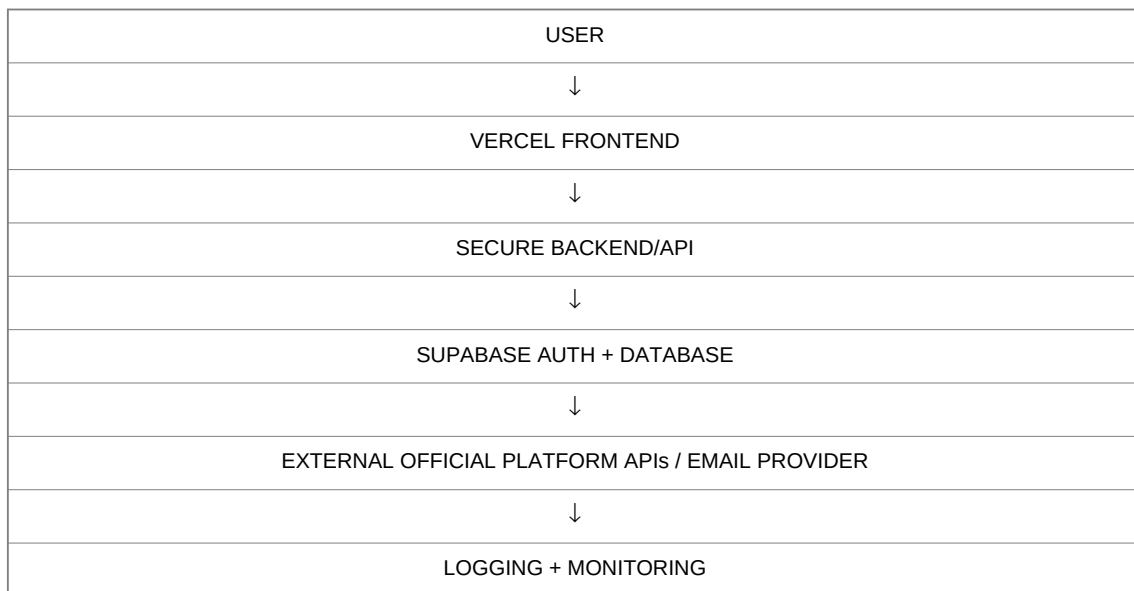
11 — Security, Testing & Production Deployment v1.0

Production readiness blueprint covering security, authentication, authorization, testing, CI/CD, Vercel deployment, Supabase production setup, secrets, monitoring, backups and release procedures.

1. Production Objective

Nibrexo OS AI ko production mein deploy karte waqt security, reliability aur recoverability ko development speed se pehle priority di jayegi. GitHub version control, Supabase backend/database, Vercel hosting and domain infrastructure core stack ka part hain.

2. Production Architecture

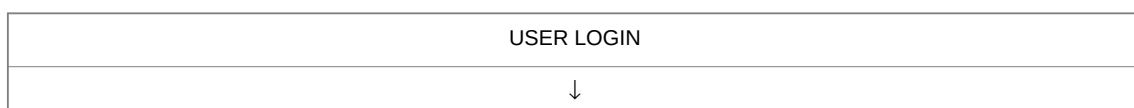


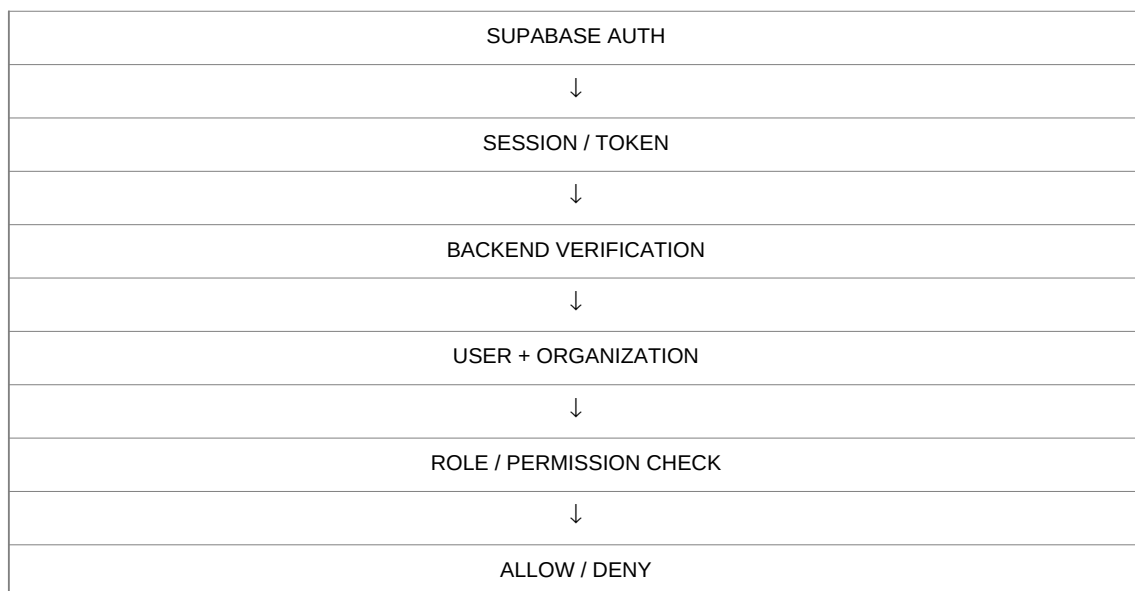
3. Environment Separation

Environment	Purpose
Local / Development	Developer testing and feature work
Preview / Staging	Pull-request and integration testing
Production	Real users and real connected accounts

- Development credentials must never be reused as production credentials.
- Production database changes require controlled migrations.
- Real client data should not be copied into development unnecessarily.
- Preview deployments must not accidentally access production secrets.

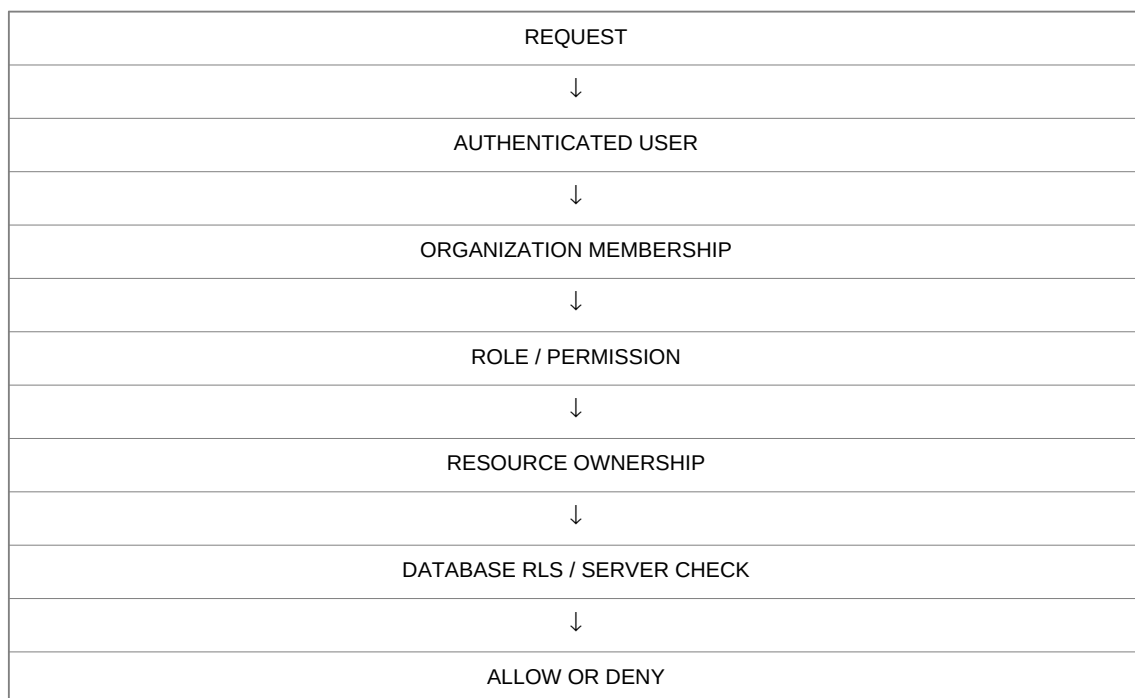
4. Authentication Security





- Protected routes require valid authentication.
- Expired sessions must be handled cleanly.
- Authentication is not authorization; permissions are checked separately.
- Account recovery and session management must use supported authentication mechanisms.

5. Authorization & Multi-Tenant Security



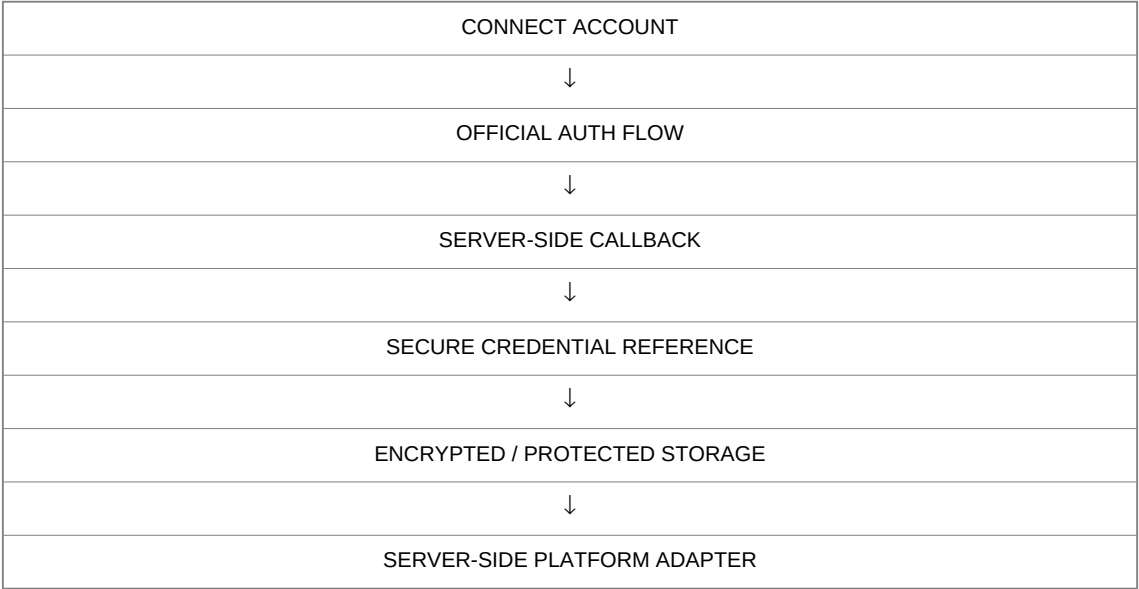
- Every organization can access only its own data.
- Supabase Row Level Security is a core database boundary.
- Server-side authorization remains authoritative.
- AI agent actions use the same permission model as normal application actions.

6. Secrets & API Keys

- Never hard-code API keys in source code.

- Never expose privileged secrets to browser JavaScript.
- Use Vercel environment variables for deployment secrets.
- Use Supabase server-side secrets only where required.
- Rotate compromised credentials immediately.
- Separate development, preview and production credentials.

7. Social & Email Credential Security

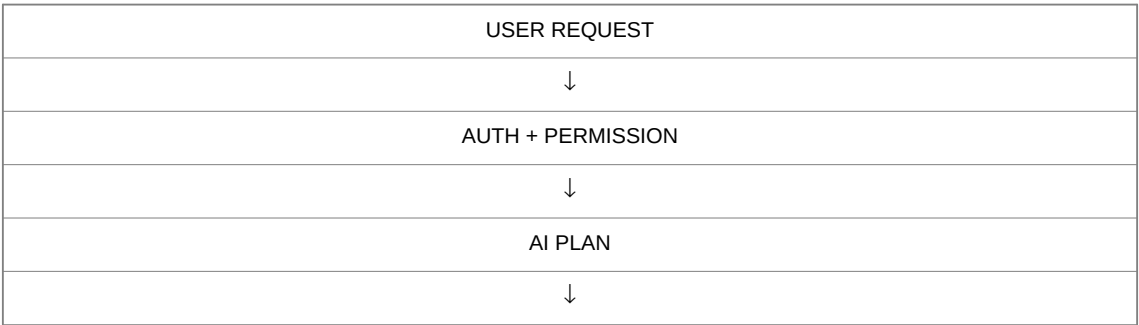


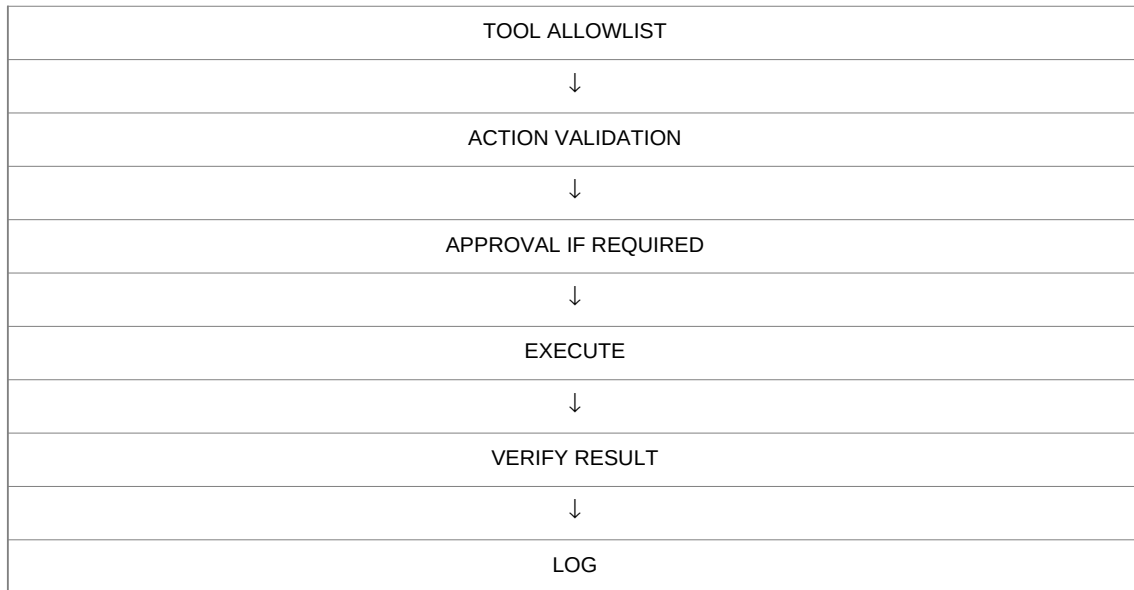
Actual storage/encryption implementation must follow the selected provider/API requirements and Supabase/Vercel security capabilities.

8. Application Security Controls

- Input validation on every API endpoint.
- Output encoding/sanitization where applicable.
- Rate limiting for sensitive endpoints.
- CSRF protection where applicable to the chosen architecture.
- Secure HTTP cookies/session handling where used.
- File upload validation for content/media.
- Strict permission checks for destructive actions.
- Audit logging for important external actions.

9. AI Security Rules



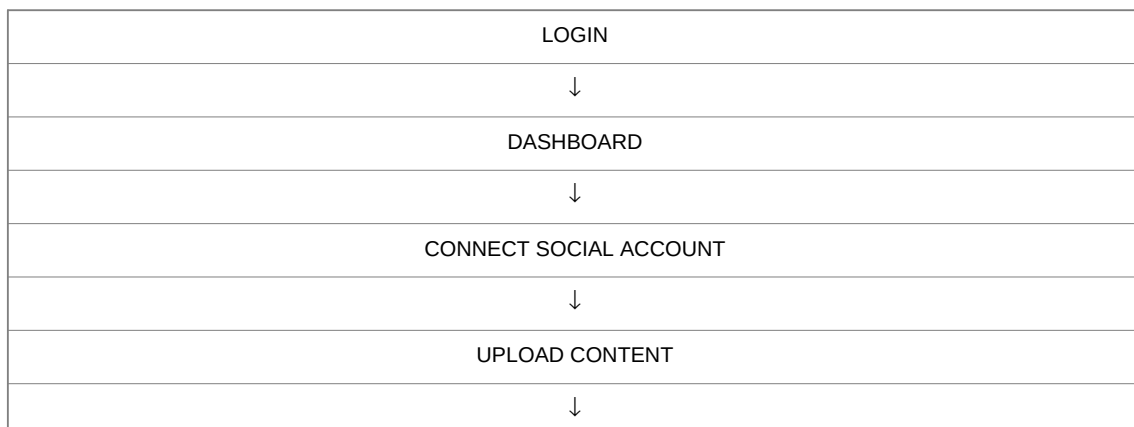


- AI cannot grant itself permissions.
- AI cannot access arbitrary tools or external services.
- AI must not claim an action succeeded without verification.
- External publishing/sending actions follow configured approval rules.
- Secrets are never included in AI prompts or responses.

10. Testing Strategy

Test Type	Purpose
Unit Tests	Individual functions/services
Integration Tests	Database/API/service interaction
API Tests	Endpoints, validation and authorization
UI Tests	Critical frontend workflows
E2E Tests	Complete user journeys
Security Tests	Auth, RLS, permissions, input handling
Regression Tests	Prevent old features from breaking
Smoke Tests	Fast production health verification

11. Critical Test Flows



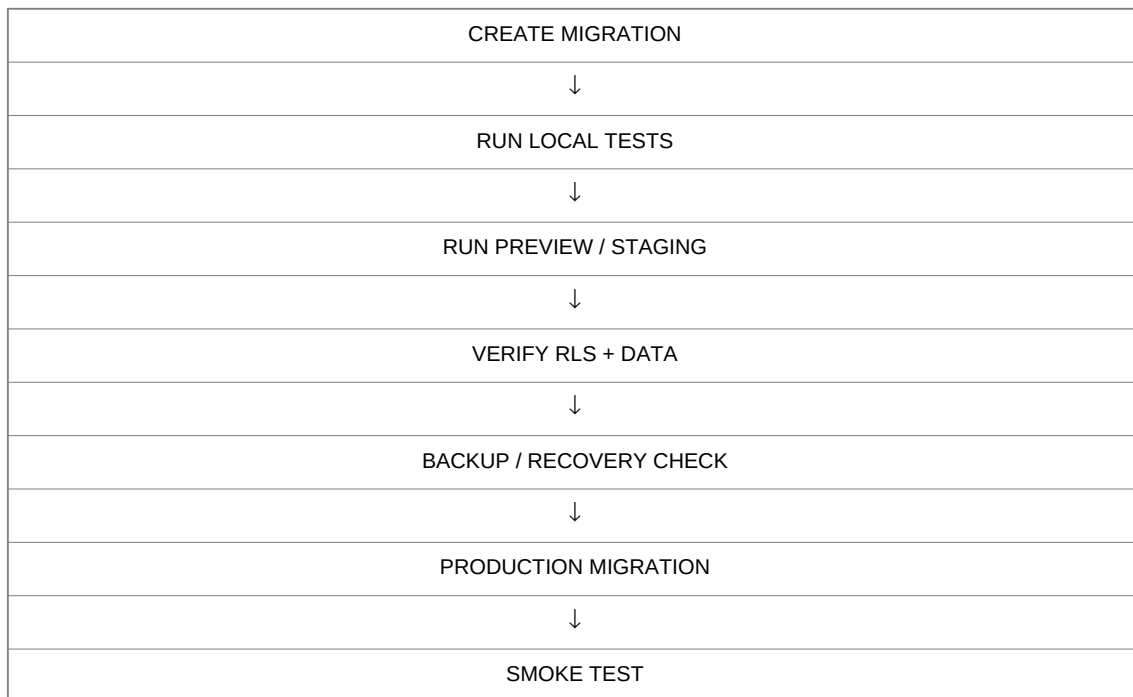


12. Negative / Failure Tests

- Invalid login/session.
- User without organization access.
- User without module permission.

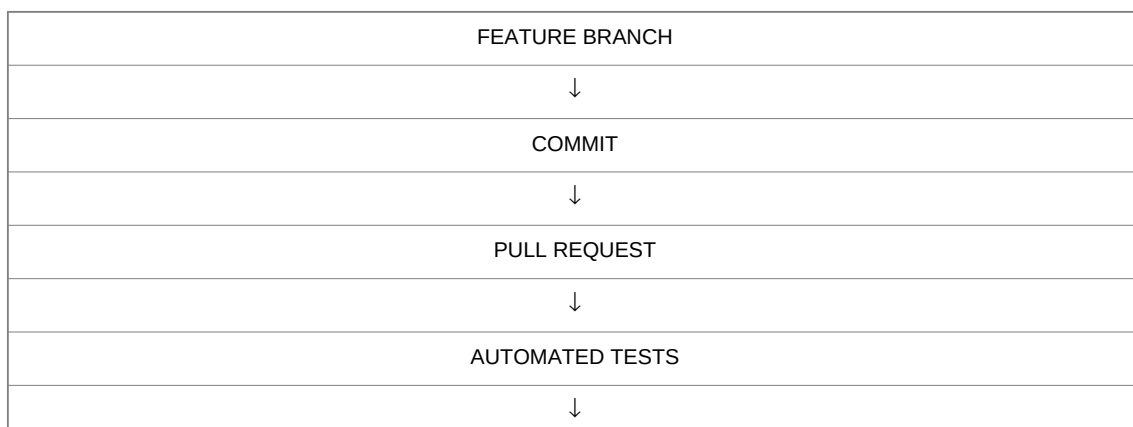
- Expired social authorization.
- Rate-limited platform API.
- Invalid media format.
- Provider email failure.
- Duplicate publish/send retry.
- Webhook delivered twice.
- Database/network timeout.
- Cancelled scheduled job.

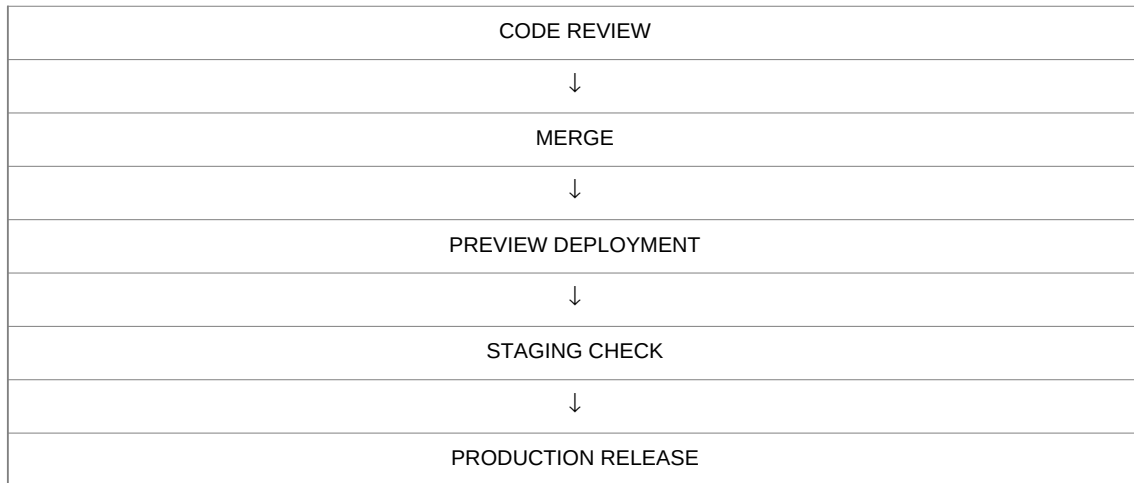
13. Database & Migration Safety



- Schema changes are version-controlled.
- Do not manually modify production schema without a documented migration.
- Destructive migrations require special review.
- RLS policies are tested after schema changes.

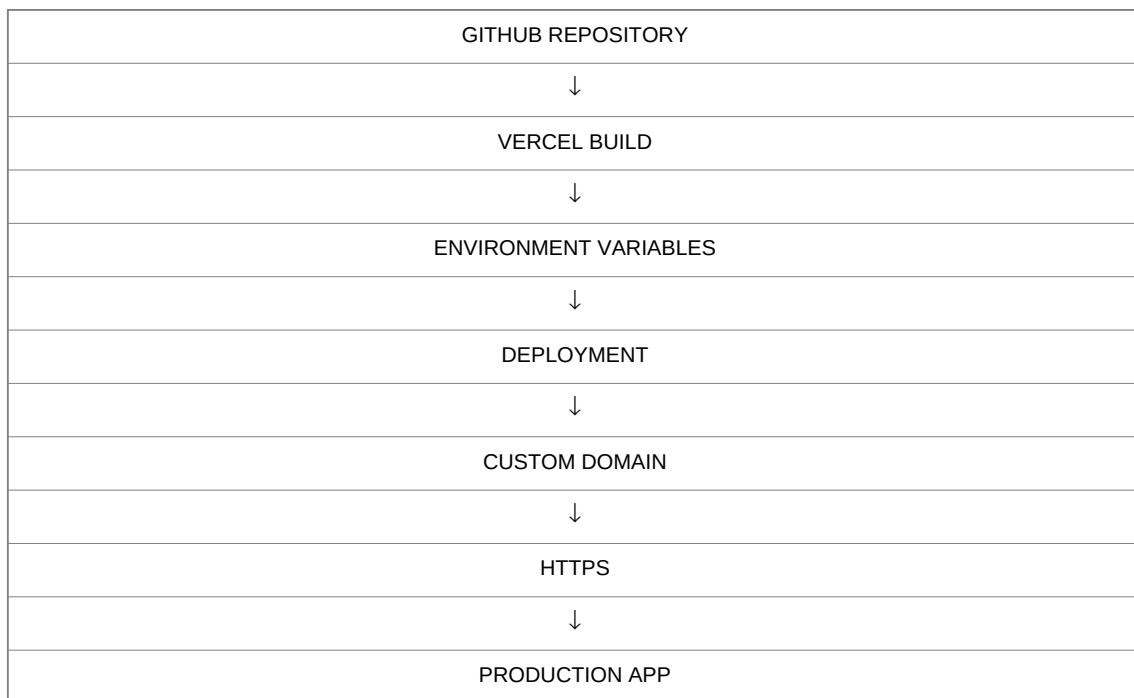
14. GitHub Development & CI Flow





- Commit messages should clearly describe changes.
- Pull requests should identify affected modules and tests.
- Failed checks block release.
- Production changes should be traceable to a Git commit.

15. Vercel Deployment Architecture



- Production and preview environment variables remain separated.
- Build must fail on critical compile/type/test errors.
- Domain configuration should use HTTPS.
- Rollback to a known-good deployment must remain possible.

16. Supabase Production Checklist

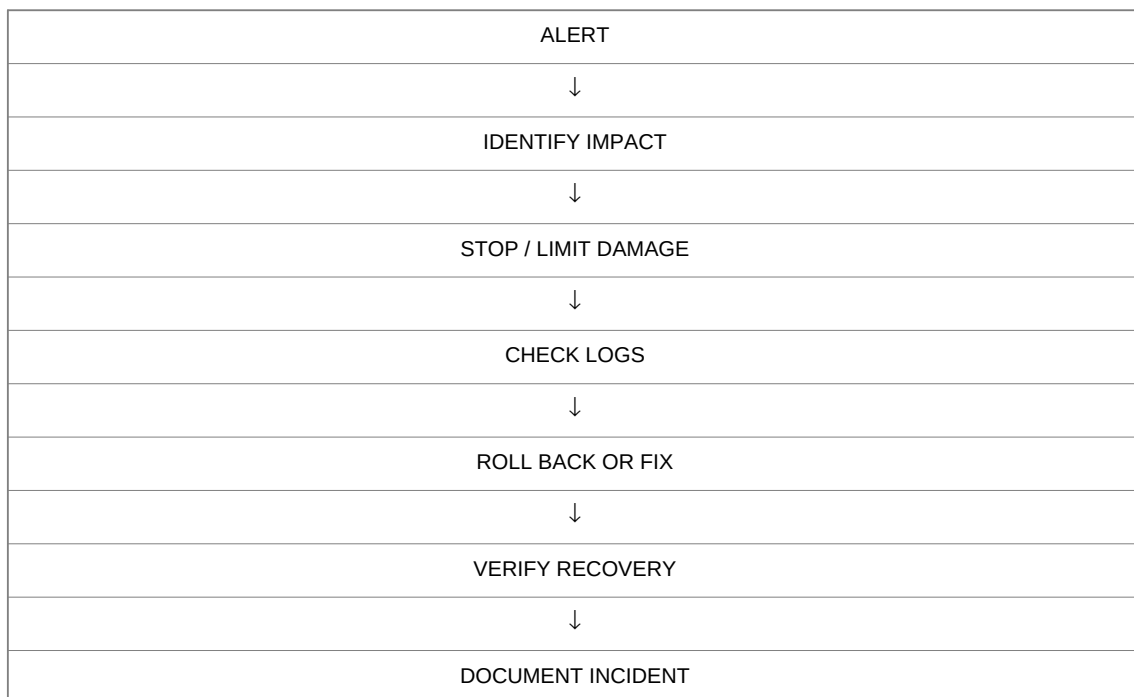
- Authentication configured.
- Database schema migrated.
- RLS enabled on organization/user data.

- Policies tested.
- Indexes reviewed for important queries.
- Storage access rules configured if media storage is used.
- Backups/recovery options reviewed.
- Production project credentials protected.

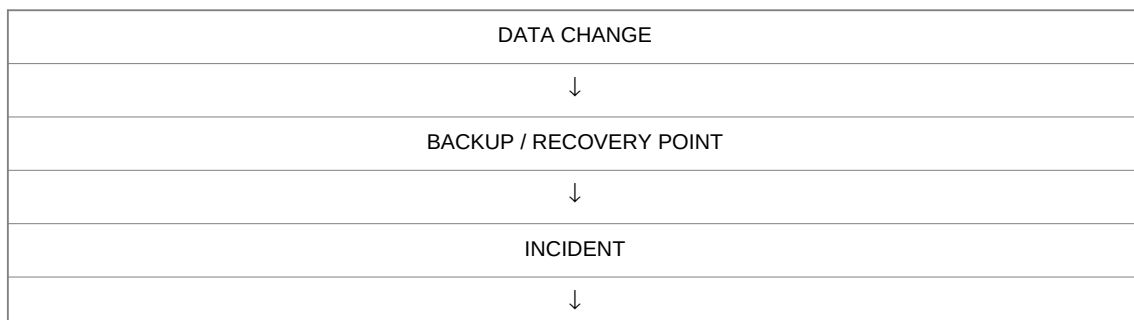
17. Monitoring & Observability

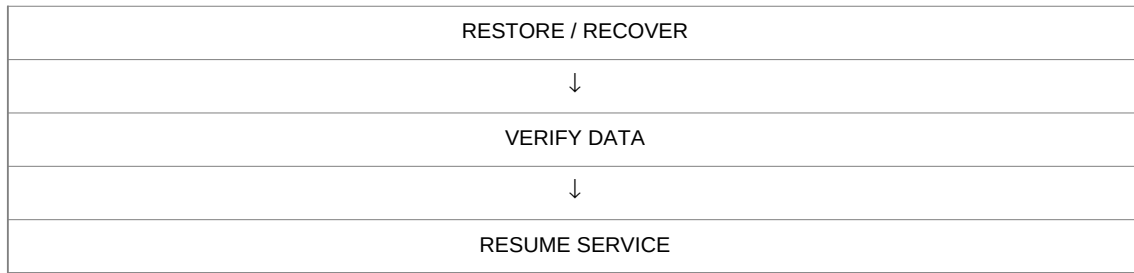
Area	Track
Application	Errors, failed requests, latency
Database	Errors, slow queries, resource health
Social	Publish failures, auth failures, rate limits
Email	Send failures, bounces, provider errors
AI	Tool failures, task failures, approval states
Jobs	Queued, running, failed, retried jobs

18. Incident Flow



19. Backup & Recovery



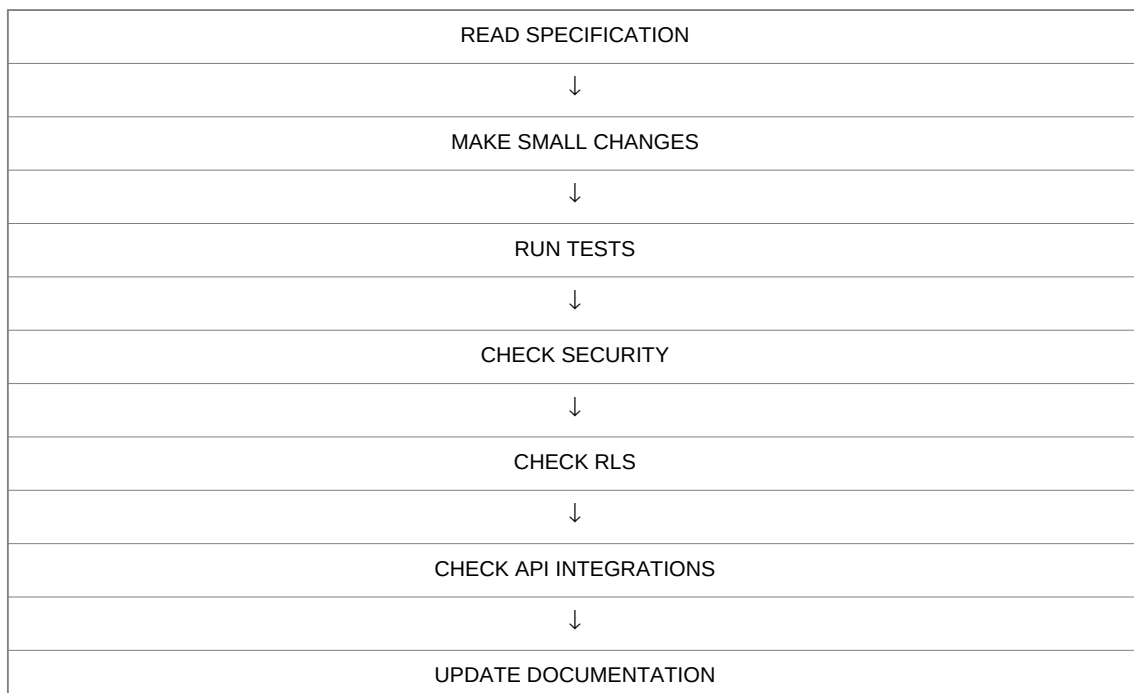


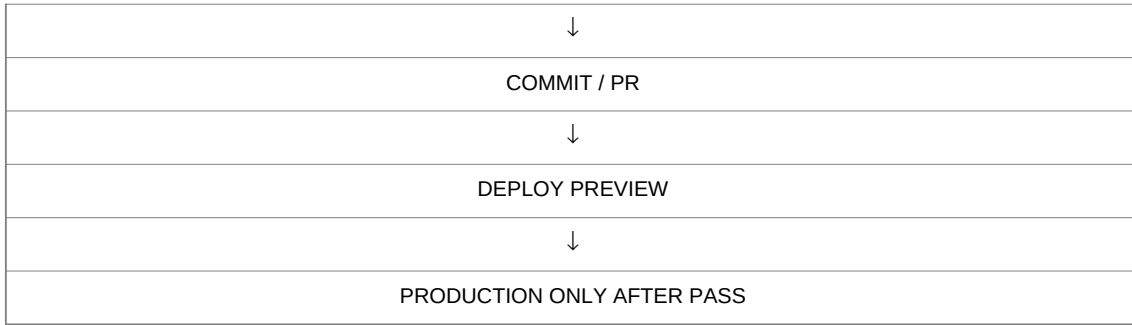
- Recovery procedures must be documented before production launch.
- Critical database data must have an appropriate backup/recovery strategy.
- Restore procedures should be tested periodically.

20. Production Release Checklist

- All critical tests passing.
- Authentication tested.
- RLS tested.
- External API integrations tested.
- Email provider tested.
- Environment variables verified.
- Database migrations reviewed.
- Production domain and HTTPS verified.
- Monitoring/error visibility available.
- Rollback plan available.
- Backup/recovery plan available.
- Final smoke test completed.

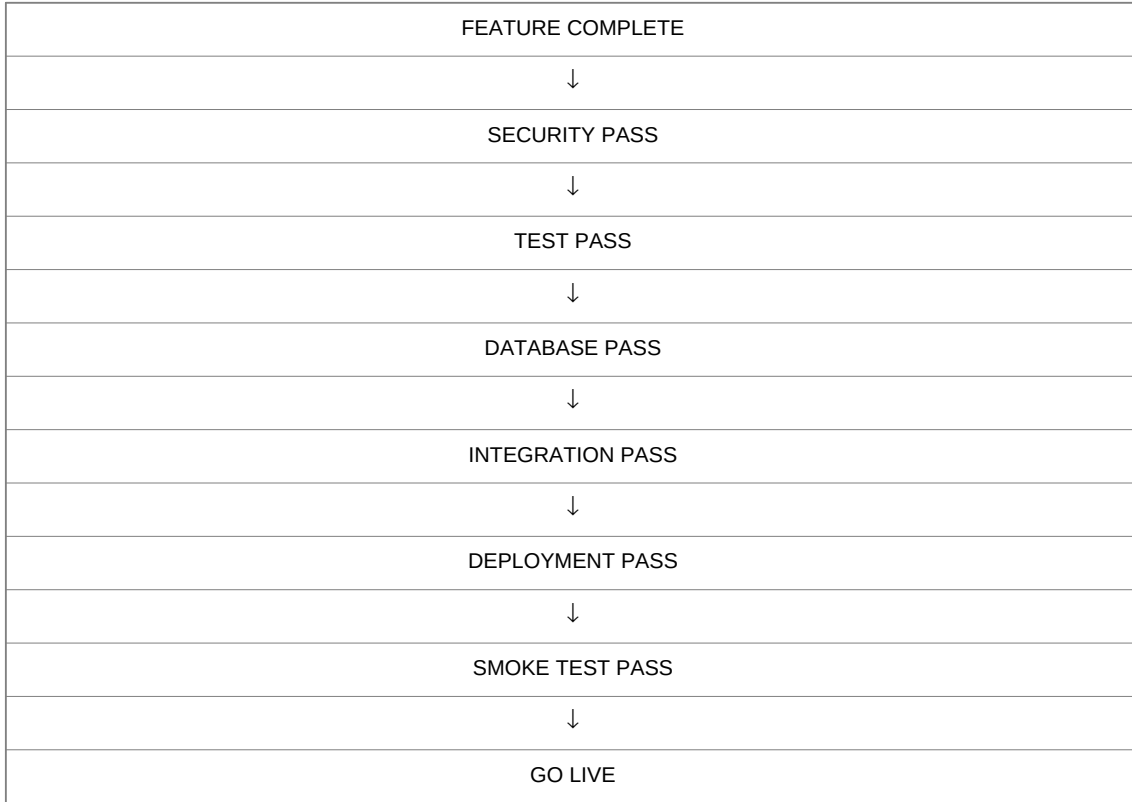
21. LLM Arena Production Coding Rules





The coding agent must not silently change production infrastructure, database permissions, environment secrets or security controls without an explicit implementation requirement and documented change.

22. Final Production Gate



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